



Forgetting other communicable diseases during the COVID-19 pandemic: Tuberculosis mortality in Peru

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Tuberculosis (TB) is currently the 13th leading cause of death worldwide and is the second most infectious killer after COVID-19. It is a disease intimately related to socioeconomic inequalities, thriving among communities where poverty and malnourishment prevail.¹ The promotion of human rights and equitable access to health care were core values of the interventions that have slowly reduced contagion and deaths due to this disease. In this context, global plans like the *End TB strategy* had set out to reduce TB mortality by 75% by 2025 and 95% by 2035.¹

The COVID-19 pandemic has undermined all progress made in this regard. Globally, healthcare utilization reduced by around a third during the pandemic.² Most countries relocated their resources to face the growing COVID-19 cases, while patients feared to approach hospitals due to risk of contagion. This is likely why the World Health Organization reported that the initial milestones for 2020 regarding the reduction of the incidence rate and mortality fell short by 9 and 25.8 percentage points, respectively. In this scenario, there was an estimated 18% decline in TB diagnosis and an increase of over 1 million deaths in 2020 compared to the previous year.¹

Peru was one of the top 16 countries that contributed to the shortfall in TB notifications. Carrillo-Larco et al. highlighted that there was a decrease in over 8000 cases of reported TB during the first months of the pandemic compared to 2019,³ suggesting an increase in transmission and complications for those affected. So far, there are no studies evaluating the impact of the COVID-19 pandemic on mortality due to TB in Peru. We conducted an interrupted time series (ITS) analysis to

assess the aggregate change in TB-related deaths before and after the mandatory lockdown in Peru in response to COVID-19 pandemics. The data was obtained from National Death Information System (SINADEF in Spanish), a national system of electronic web registration of deaths implemented in 2016.⁴ After the mandatory lockdown, we found an average increase of 3.3 (95% C.I: 1.29 to 5.33) TB-related deaths. This increase was far more noticeable among men compared to women (2.01 versus 1.29) and is in line with the global trend estimated until 2020⁵ (Figure 1).

The COVID-19 pandemic has disrupted the systems designed to manage communicable and non-communicable diseases. Even though a high priority has been given to diseases such as cancer, mental illnesses, and strokes, TB should also be treated with urgency to prevent unnecessary deaths among vulnerable people in low resource settings and a loss of productivity among society at large.⁶ A call to action should be made on increasing awareness of TB at both the population and government level. Particularly in the region of America, which suffers from disjointed and underfunded health systems. Increased financial support and solid public policies must be implemented to prevent the resurgence of this disease and get back on track to meet the End TB plan.

Contributors

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Declaration of interests

The authors declare no competing conflict interests.

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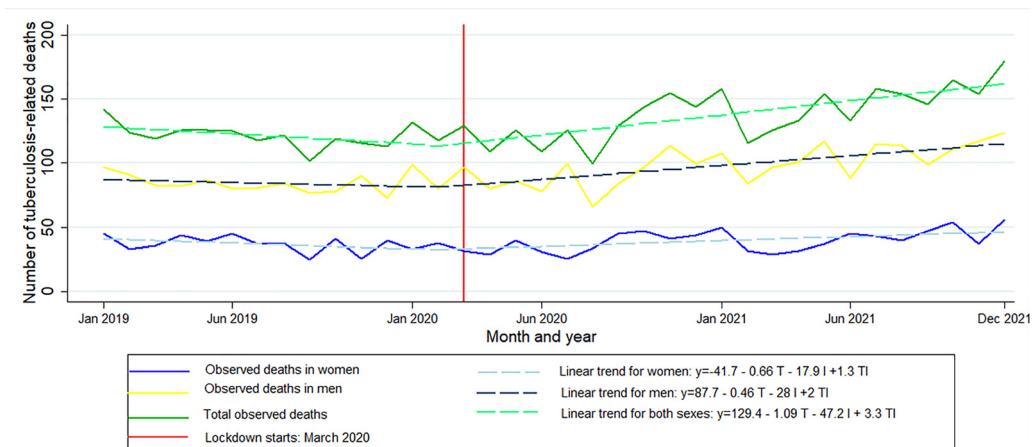


Figure 1. Interrupted time series analysis of tuberculosis deaths by sex in Peru: 2019–2021. Source. National Peruvian Death Registry (SINADEF). ICD-10 codes A15.0 to A19.9 were used to define TB cases. The intervention variable takes the value of one after implementing the mandatory lockdown in response to COVID-19 pandemics in Peru (from March 2020 to December 2021), and zero otherwise (from January 2019 to February 2020). We used the following equation: $Y_T = \beta_0 + \beta_1 T_t + \beta_2 I_t + \beta_3 TI_t + \mu_t$, in which Y_T is the aggregate outcome variable, T_t is the time, measured every month that has passed since the start of the study, I_t is the intervention variable and TI_t is an interaction term, β_0 is the mean value of the outcome of interest at the start of the study, β_1 is the value of the slope of the outcome of interest, β_2 is the change in the slope of the outcome of interest immediately after the intervention, β_3 is the difference of the slope after and before the intervention and μ_t represents the stochastic term of the statistical model.

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